

Your class played **Earn to Fire** with InfoSec Battle Bots at STEMfest Newcastle. In two teams, students answered STEM questions to fire real hazards in a robot combat arena while classmates drove the robots. Every wrong answer came with an explanation.

Topics covered (KS3)

Forces & levers	Fulcrums, Newton's laws, weight = mass $\times$ g, centre of gravity
Circuits & electricity	Ohm's law ($V = I \times R$), $P = V \times I$, series circuits, fuses, solenoids
Gears & motion	Speed = distance / time, RPM, gear ratios, circumference
Code & logic	Variables, loops, IF statements, binary, AND / NOT
Sensors & robot brains	Ultrasonic ranging, inputs vs outputs, actuators, microcontrollers

Back in class: five follow-up questions

1. A spinner weapon spins **twice as fast**. Why does it hit **four times** harder? ($KE = \frac{1}{2}mv^2$)
2. A 12 V battery drives a 4 Ω motor. What current flows, and how much power does it use?
3. Design a pit-avoiding robot: which sensor would you use, and what would your **IF** statement say?
4. Why do battle robots use a **low centre of gravity**? Sketch a robot that is hard to flip.
5. Count to 31 in binary on one hand. How high can you count with two hands?

Answers: (1) energy grows with speed squared, so $2^2 = 4$; (2) 3 A, 36 W; (3) e.g. a downward light or ultrasonic sensor: IF floor not detected THEN reverse; (4) it must be lifted further before it tips; (5) 1023 (10 bits).

How the arena works

The iPads talk over WiFi to a Windows server running a Node.js program. It checks each answer, then sends commands to a **Raspberry Pi**, which switches the arena's motors, solenoids and servos. It is real networking, coding and electronics: a good talking point for careers in engineering and cyber security.

Want more? Termly schools competitions

We're building towards a schools competition every term, working with local **STEM Ambassadors**. Our schools work is led by **Lewis Terry**. Get in touch to take part:

Email	info@infosecbattlebots.com (subject "Schools")
Schools page	www.infosecbattlebots.com/schools
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The game's code	github.com/Ben-PyroGuy-Doch/IBB-Arena-STEM-Challenge (open source)